ANNDA – Lab assignment 1

Classification with a single-layer perceptron

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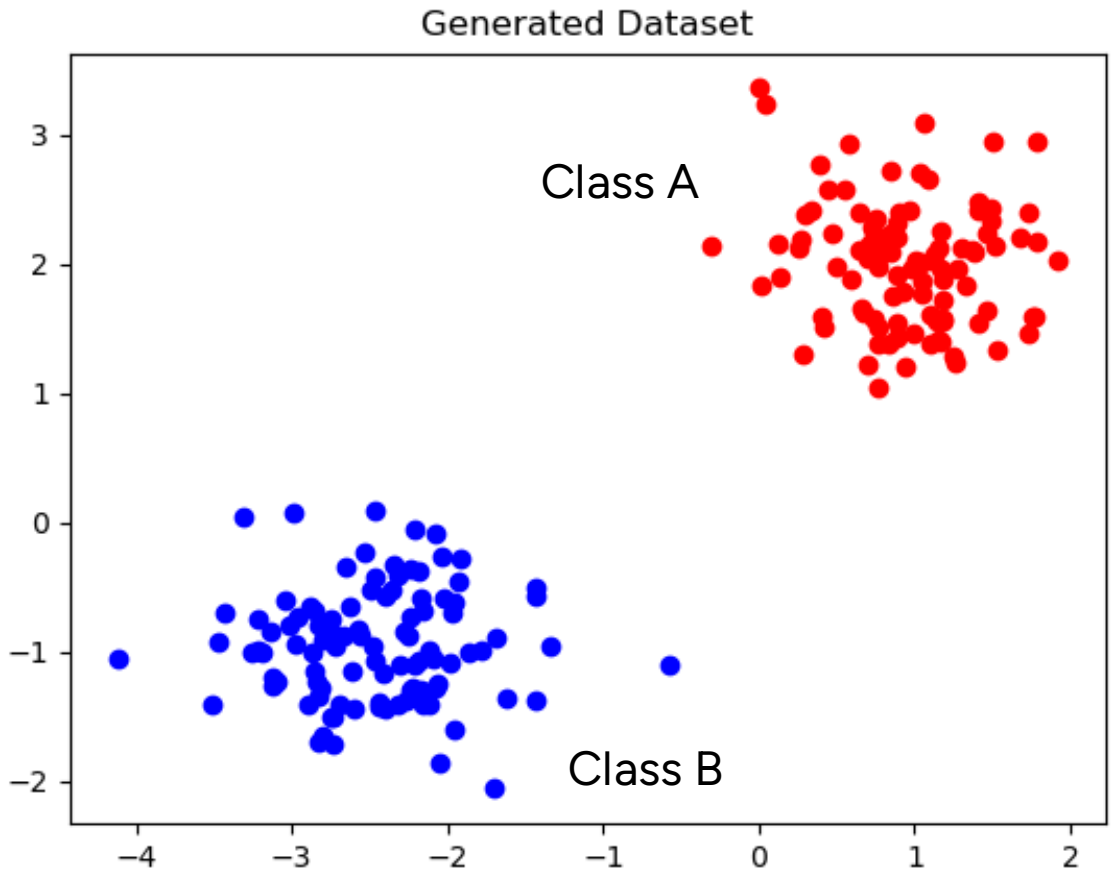
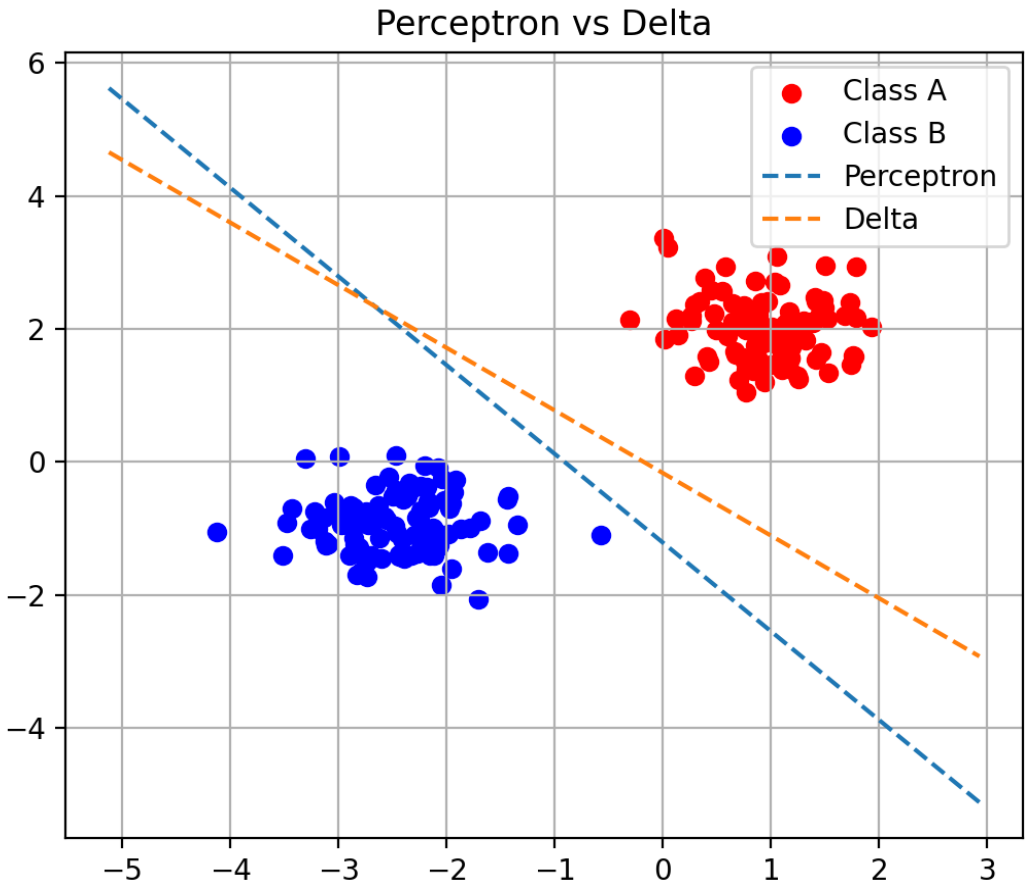
3.1.1. Generation of linearly-separable data



Class A
m=(1, 2)
sigma=0.5

Class B
m=(-2.5, -1)
sigma=0.5

3.1.2. Classification with a single-layer perceptron

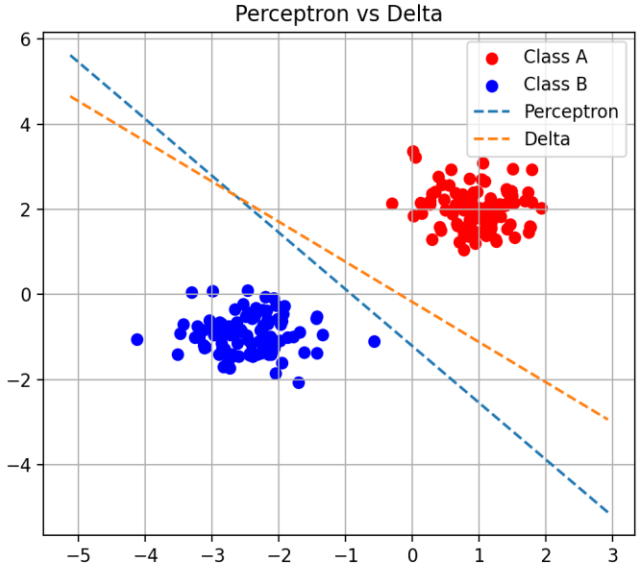


Convergence

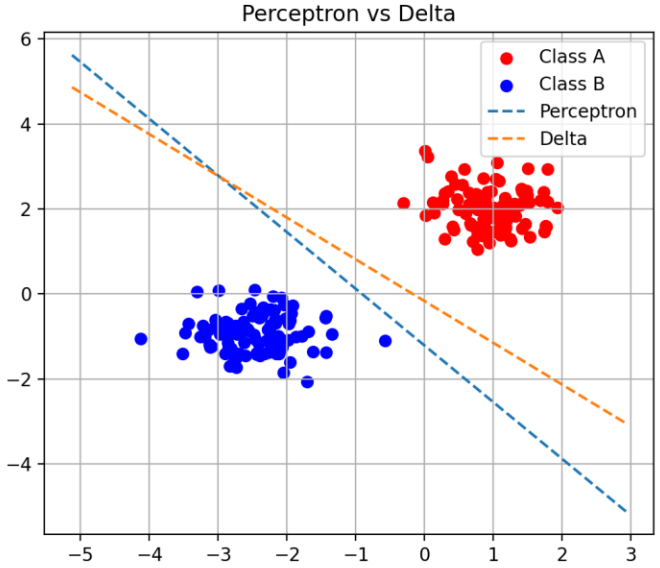
- Both approaches converge
- Delta learning creates a boundary that seems more appropriate between the two classes; it is approximately the same distance from both classes.
- Classical perceptron has one sample very close to the boundary

3.1.2. Classification with a single-layer perceptron

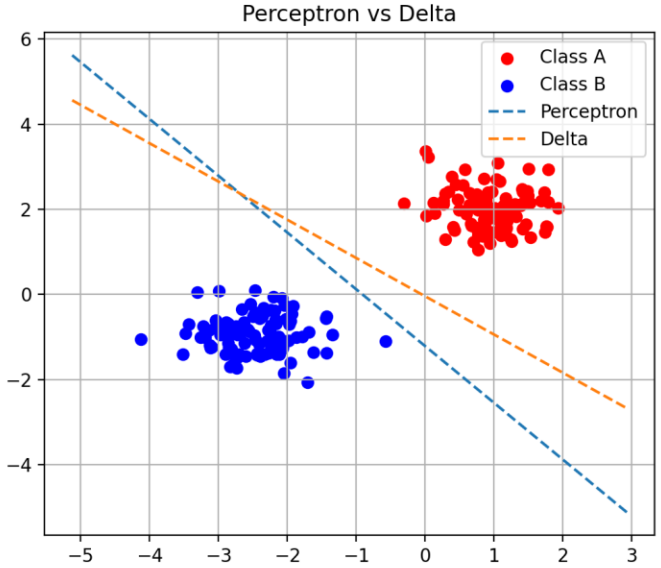
lr=0.001



lr=0.01



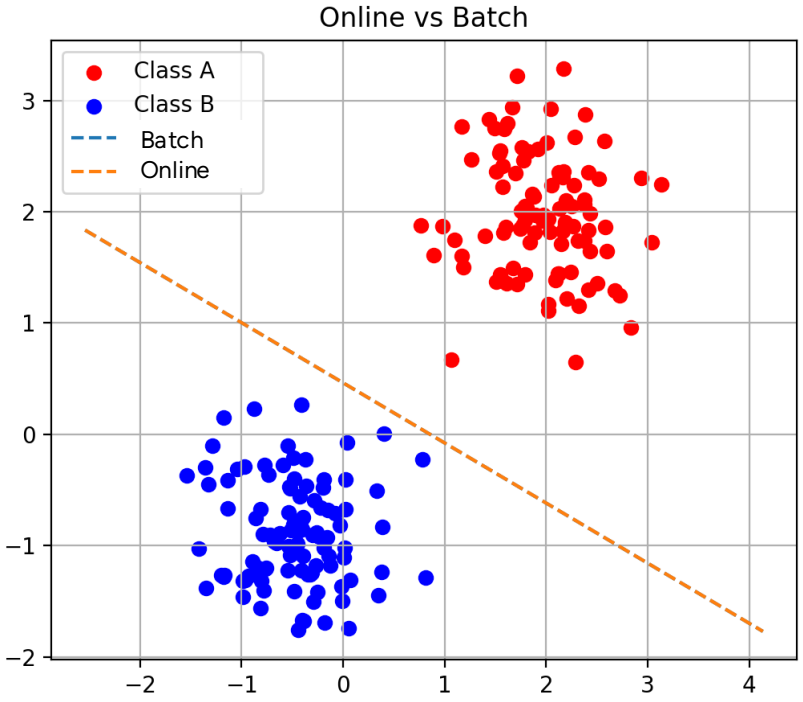
lr=0.1

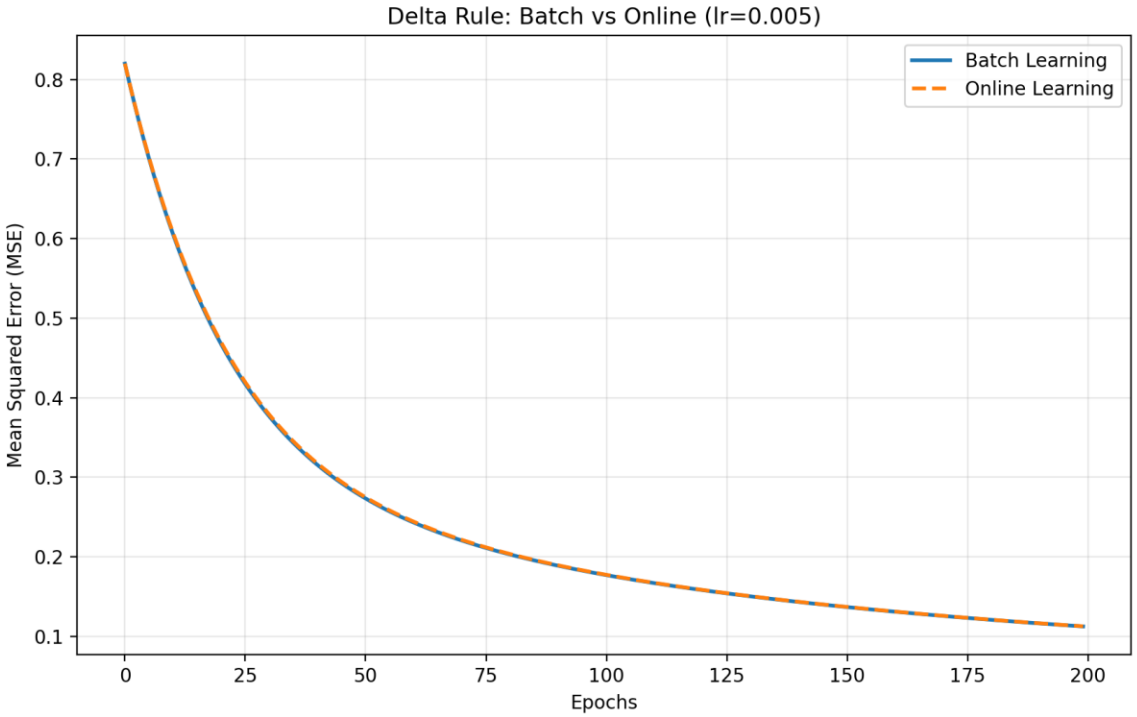


Learning rate

Changing the learning rate doesn't affect the result of classical perceptron learning but have impact on delta rule

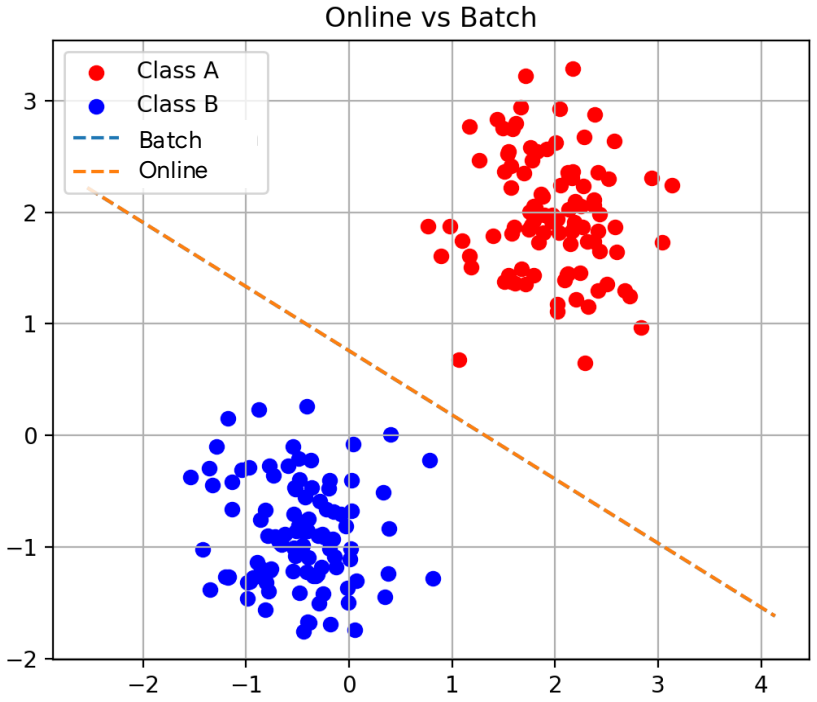
3.1.2. Classification with delta rule: online vs batch

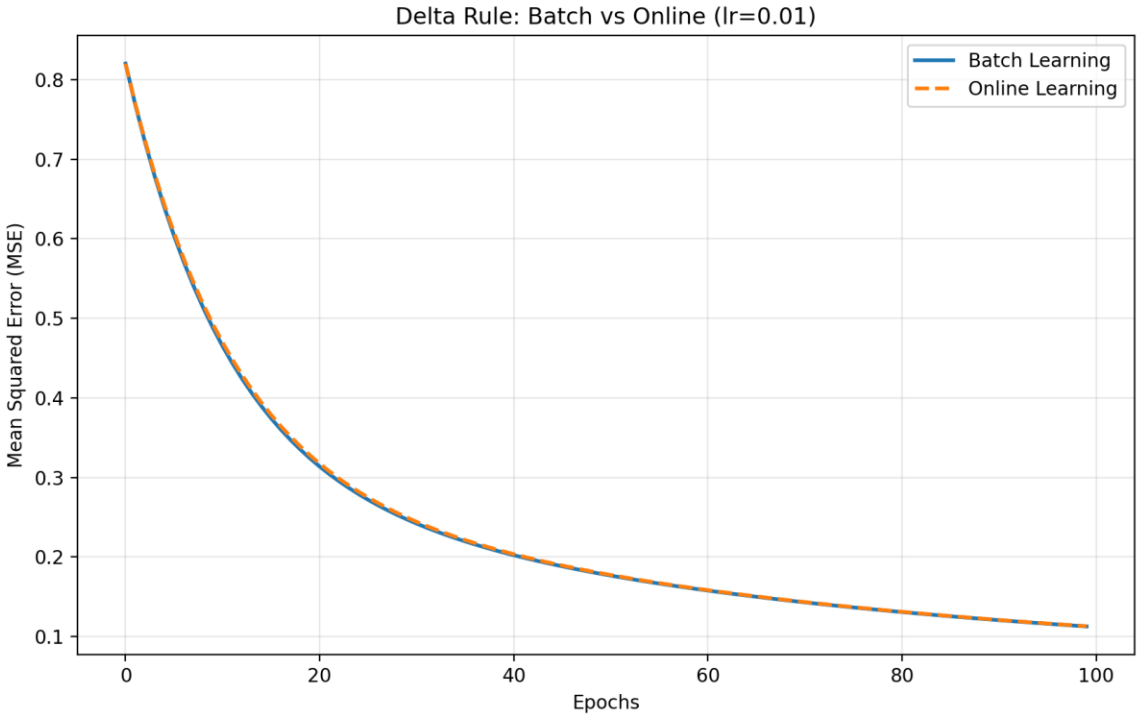




Learning rate = 0.005

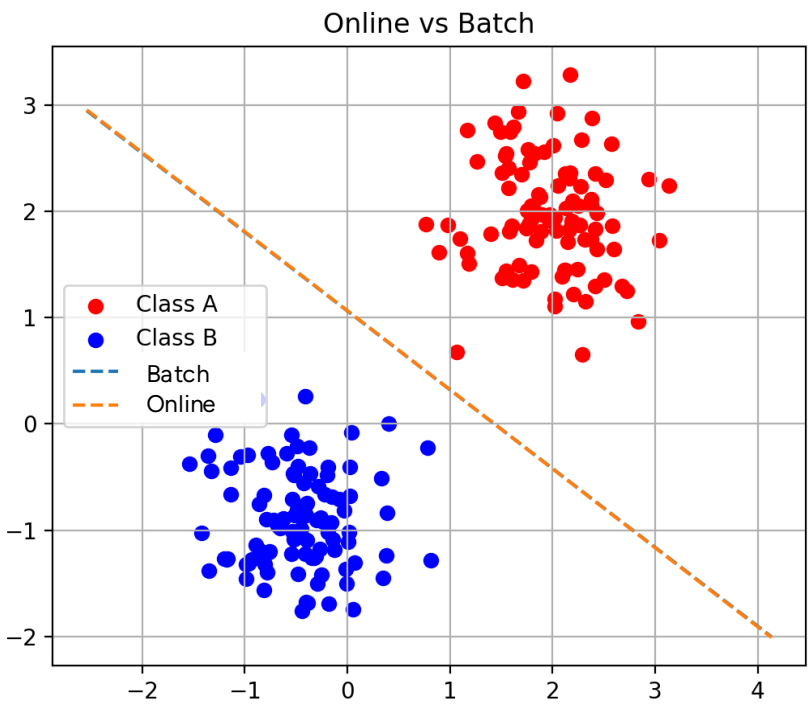
3.1.2. Classification with delta rule: online vs batch

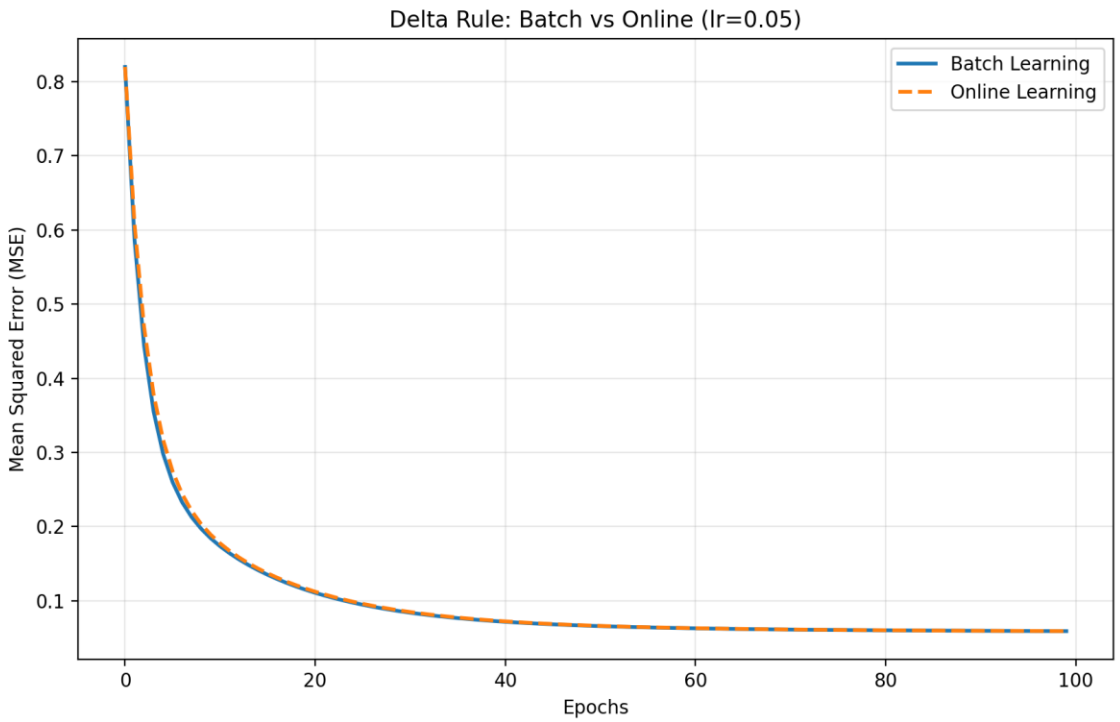




Learning rate = 0.01

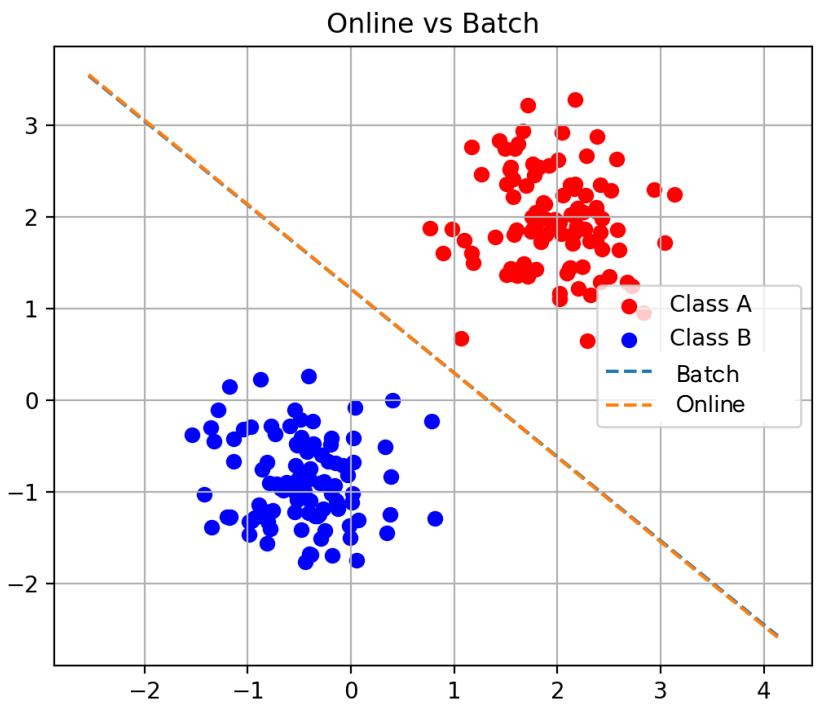
3.1.2. Classification with delta rule: online vs batch

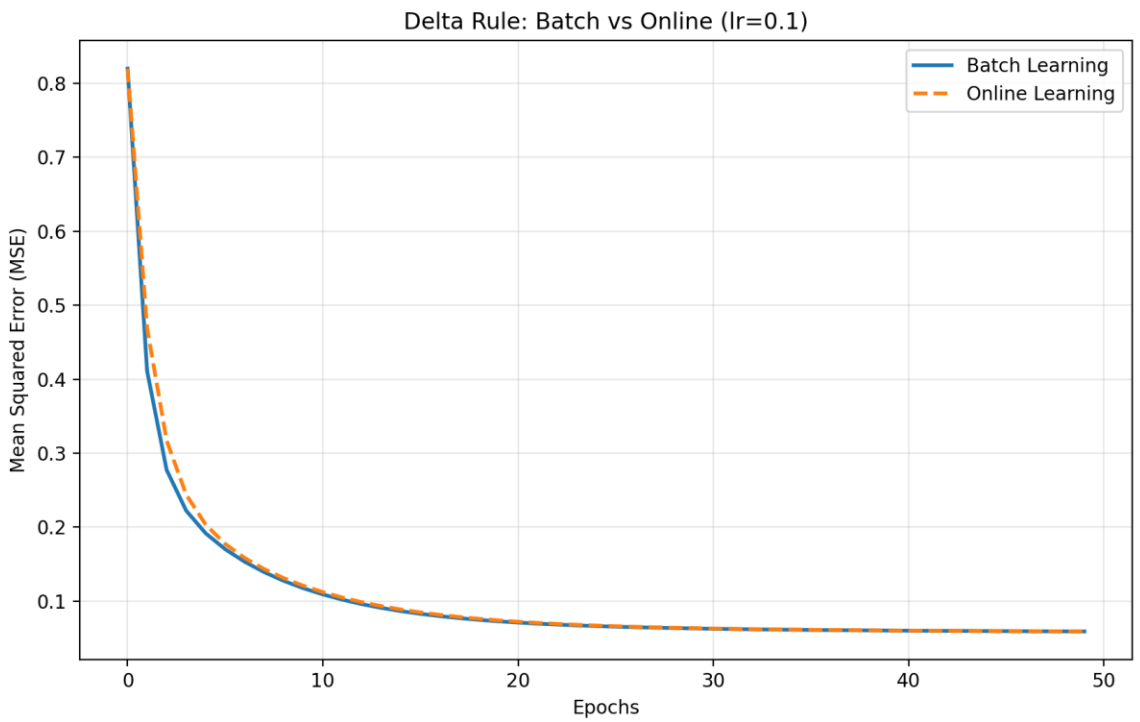




Learning rate = 0.05

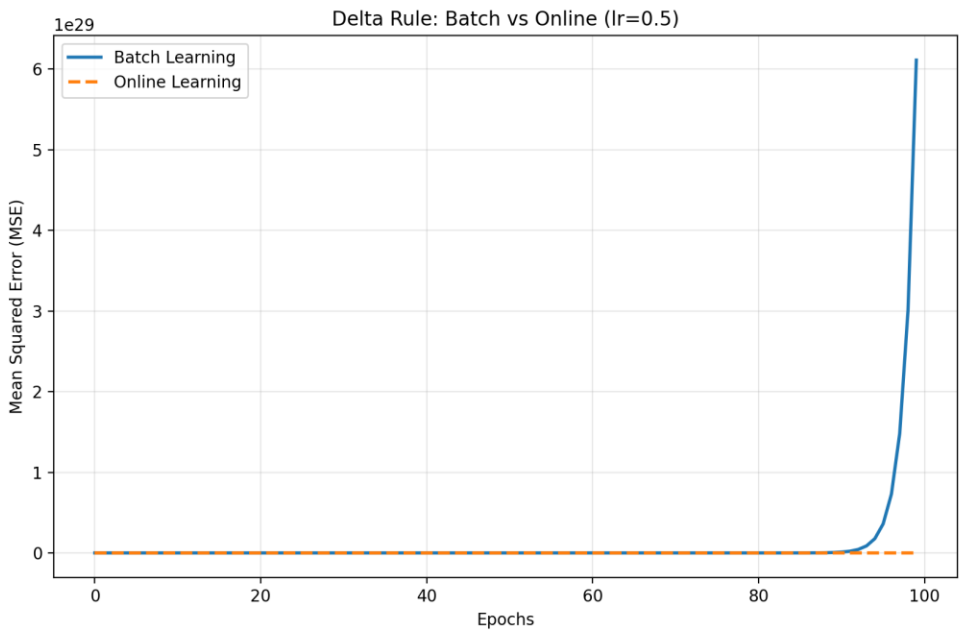
3.1.2. Classification with delta rule: online vs batch





Learning rate = 0.1

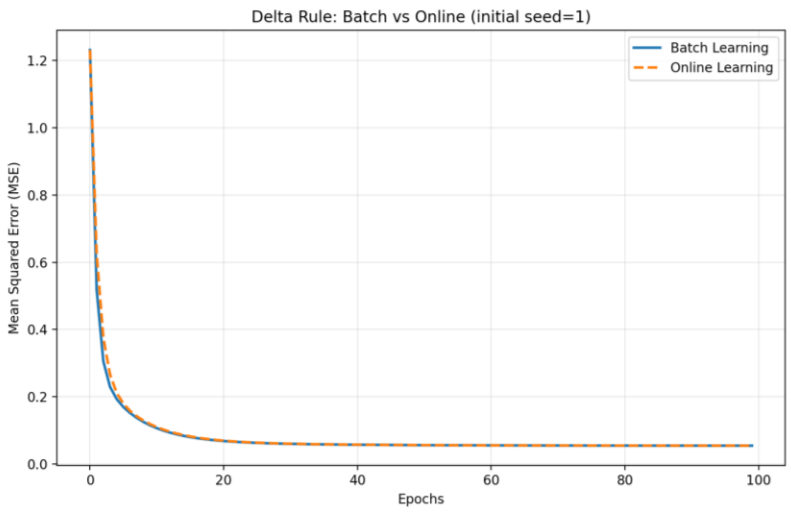
3.1.2. Classification with delta rule: online vs batch

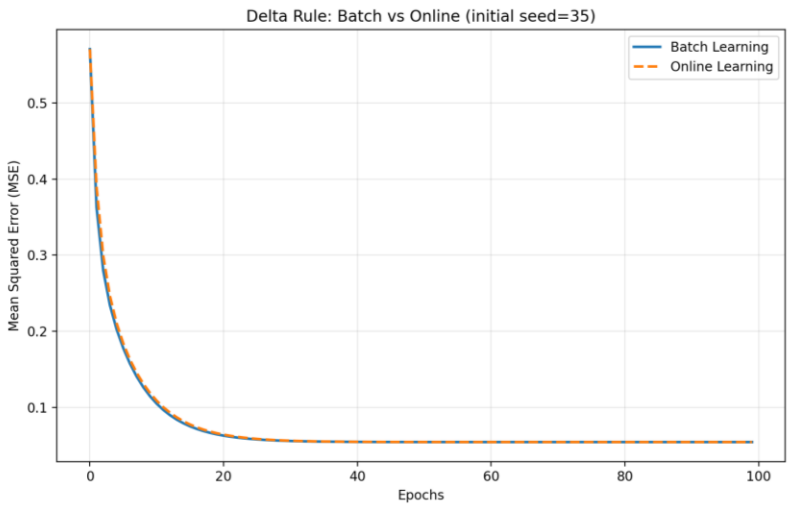


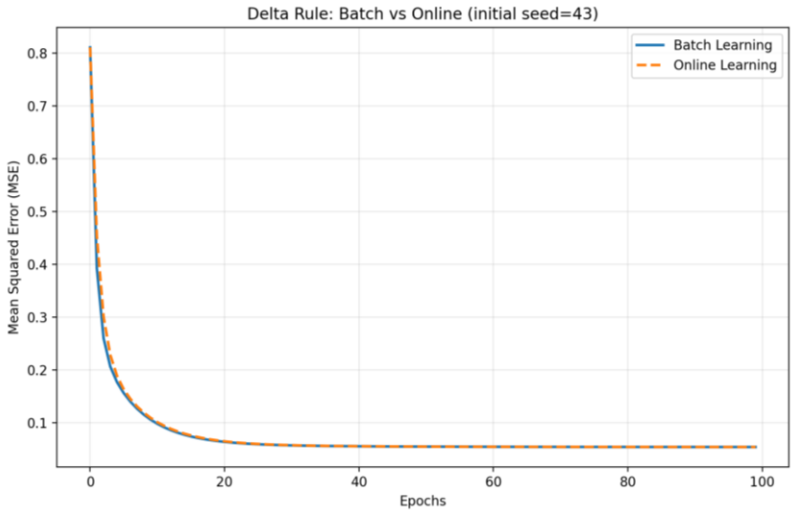
Analysis:

- Because the task is simple, there is almost no difference between the online and batch methods. However, it can still be seen that the online method converges faster in the early stages.(as shown in picture when lr =0.1)
- The larger the learning rate, the faster the convergence tends to be. But when the lr becomes too large (eg. 0.5), the updates can overshoot, causing excessively large jumps in the weights for the delta rule (batch) method, and leads to significant deviations and prevents the algorithm from converging.

3.1.2. Classification with delta rule: Different initial conditions

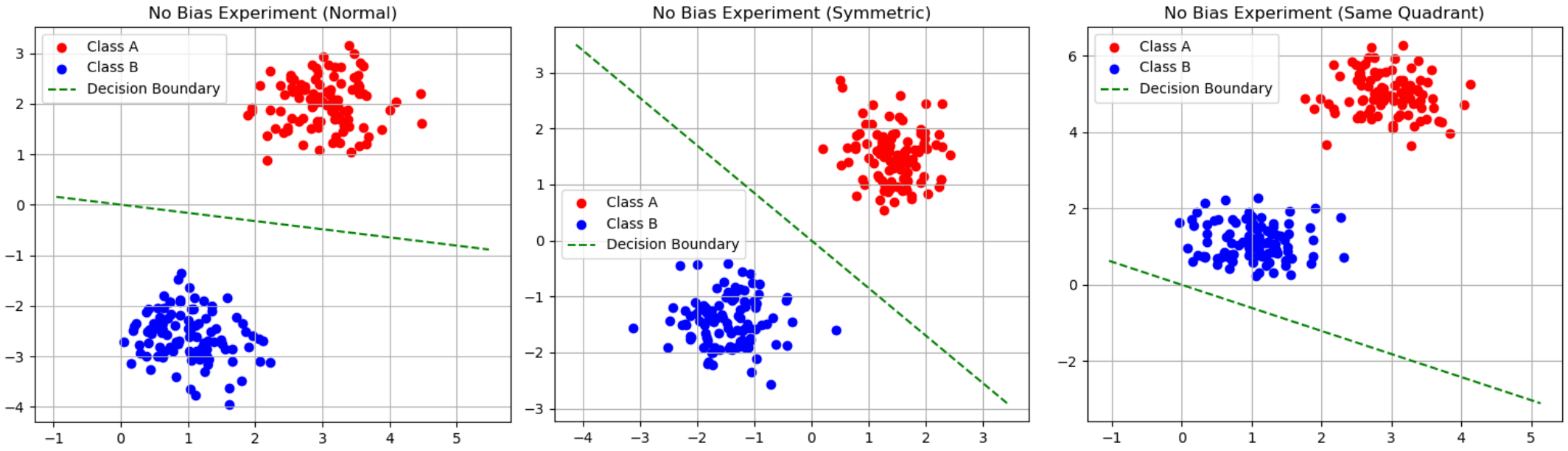






- Different initial conditions can lead to different convergence speeds, but the effect is not significant.

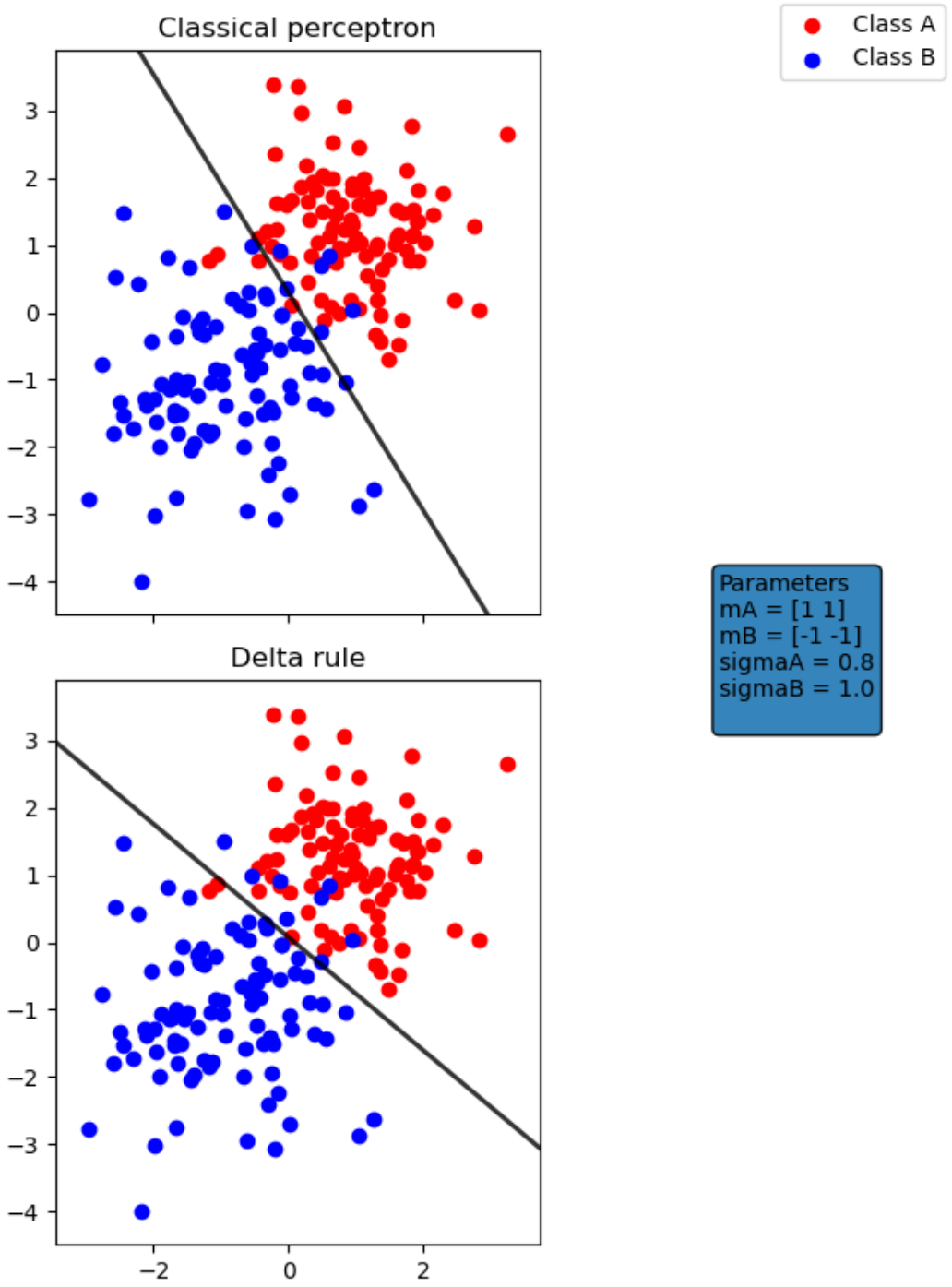
3.1.2. Classification with delta rule: Bias

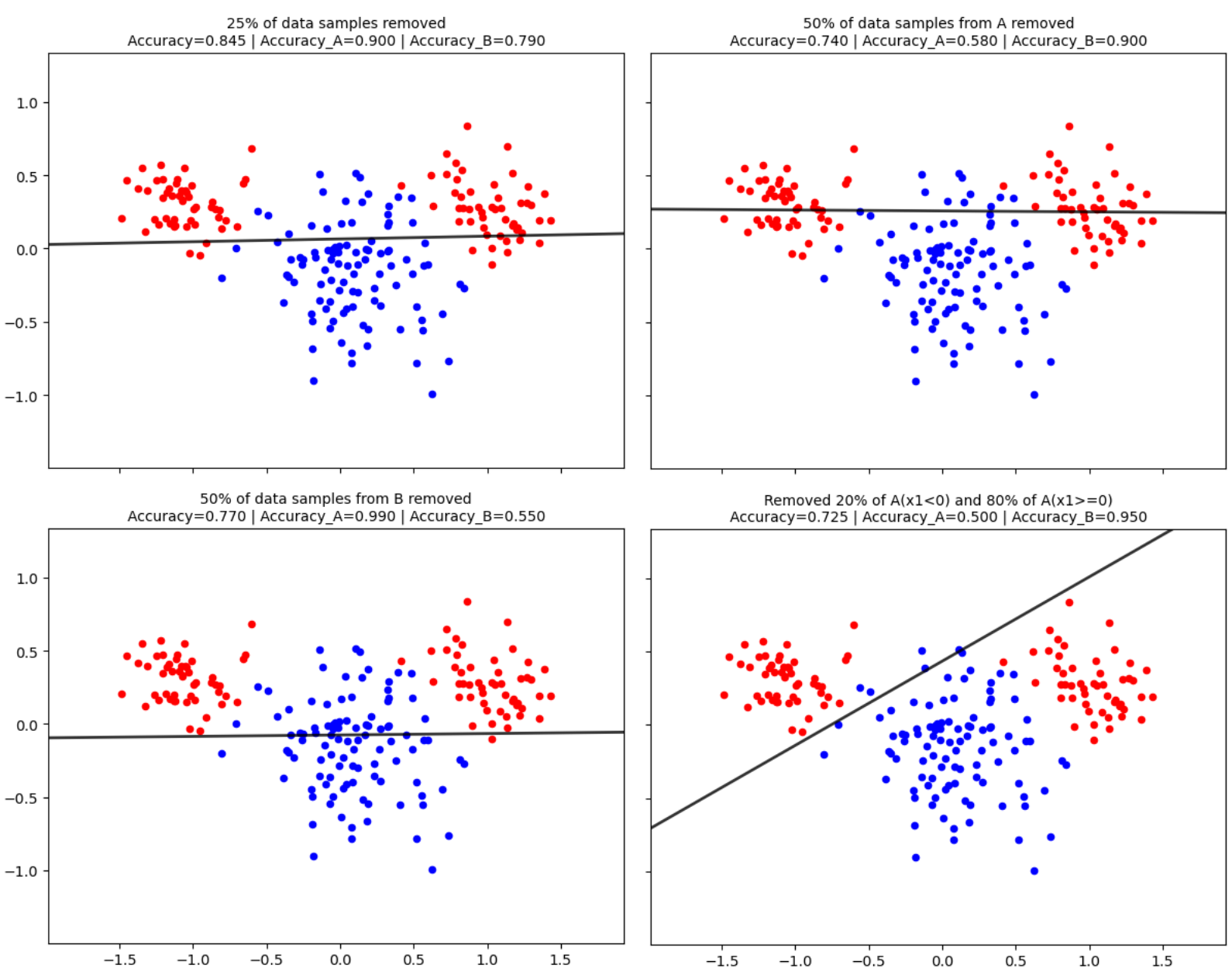


- 2 classes within the same quadrant, or 2 classes collectively spreading across 3 quadrants - **correct classification not possible without bias term.**
- 2 classes symmetric with respect to the axes or the origin, or all data points lying in 2 different quadrants - **correct classification achieved without bias term**

3.1.3. Classification of not linearly separable samples

- Not every sample of every class is classified correctly
- Delta rule seems to classify better class A, while class B is better classified by classical perceptron
- It is hard to tell which method performs better since the data is overlapping which means that the class corresponding to certain area can be ambiguous.



3.1.3. Classification of not linearly separable samples

- The boundary is moved and favor the class with more samples.
- Inside the same class we can see that the boundary is biased towards the subset with more data points.
- **A representative dataset is essential for reliable classification.**